LTRC 2.0 • GLOBAL STANDARD FOR INDIC AI

SLT-Eval 2027

Speech and Language Technology Evaluation Workshop

Evaluation as a First-Class Research Problem for Indian Languages

Progress Report & Forward Plan

Language Technologies Research Centre • IIIT Hyderabad

Agenda

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Why SLT-Eval; positioning within LTRC 2.0 and Global Standard for Indic AI

02 Workshop Goals

What the workshop sets out to standardise for Indian speech and language tech

03 Call for Papers

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04 Shared Tasks

Speech, Language and Data challenges planned so far

05 Host Body Trials

IEEE SPS vs ISCA (status and tradeoffs)

06 Forward Plan

Success criteria, immediate next steps, unresolved risks

Why SLT-Eval

The evaluation gap that holds Indic AI back

The Problem

- Indian ASR, TTS, MT, LLMs and S2S systems are each measured on their own test sets, with their own metrics, in their own format
- LibriSpeech, FLORES, CommonVoice were never built for code-switching, script diversity, dialectal variation or noisy mobile-first audio
- Progress in Indic speech and language looks slower than it actually is, because there is no common ground to demonstrate it on
- WER, BLEU and MOS have become proxy optimisation targets instead of measures of real-world usability

SLT-Eval's Response

- One umbrella covering ASR, TTS, MT, S2S, Spoken QA, Speech LLMs and rich-data tracks under a single evaluation workshop
- Benchmarks designed ground-up for Indian conditions (code-switching, script diversity, dialectal variation, noisy audio, etc.)
- New evaluation measures that go beyond WER, BLEU and MOS, with statistical rigour and reference-free options
- Coverage that extends to all low-resource and underserved Indian languages, not only the top five or ten

Positioning in LTRC 2.0

Where SLT-Eval sits in the Global Standard for Indic AI initiative

BUILD

Indic Foundation Models

Speech and language models built by LTRC and the wider research community across ASR, TTS, MT, LLMs and Speech LLMs

MEASURE

SLT-Eval

Standardised, linguistically grounded evaluation across modalities (benchmarks, metrics, leaderboards, shared tasks)

DEPLOY

Downstream Usecases

Real-world deployment of Indic ASR, TTS and MT into education, healthcare, governance and accessibility with robust benchmarks

Workshop Goals

What SLT-Eval 2027 sets out to standardise

01
Robust Benchmarks

Comprehensive benchmarks covering ASR, TTS, MT, Indic LLMs, Speech LLMs, S2S and Spoken QA across various dimensions.

02
Novel Evaluation Metrics

Metrics for TTS, ASR and S2S that go beyond WER, BLEU and MOS Measures that capture intent, prosody and Indic morphological richness.

03
Ground-Up Indic Design

Benchmarks built for code-switching, dialectal variation, script diversity, schwa deletion and noisy real-world audio, not retrofitted from English.

04
Low-Resource Coverage

Evaluation infrastructure for languages beyond the top five or ten, including Dravidian and tribal-script languages.

05
Reproducible Comparison

A standardised, high-quality evaluation foundation that researchers, startups and labs can adopt as the common standard.

06
Evaluation as Research

Reframe evaluation itself as a first-class research problem with statistical rigour, confidence intervals and reference-free metrics.

Call for Papers

Topics span speech, language and multimodal evaluation

SPEECH TRACKS

- Evaluation of ASR
- Evaluation of TTS
- Speech Translation evaluation
- Speech-to-Speech systems
- Speech LLM evaluation
- Expressive and prosodic speech
- Evaluation metrics beyond WER
- Bias and robustness in speech
- Spoken Language Cognition

LANGUAGE TRACKS

- Evaluation of MT
- Evaluation of NLG
- Evaluation of LLMs
- Question Answering evaluation
- Summarization faithfulness
- Dialogue and conversational systems
- Cross-lingual and code-mixed
- Multimodal (speech–text–vision)
- Hallucination detection and factual consistency

Shared Tasks: Speech

Twenty challenges across ASR, TTS, S2S, Spoken QA, Speech LLMs and Data

ASR

5 challenges

- Evaluation measures beyond WER
- Code-switched ASR (Hindi-English, Tamil-English, etc.)
- Domain-specific ASR (agriculture, health, legal)
- Low-resource ASR — doing more with less
- Long-form ASR (lectures, meetings, court audio)

TTS

4 challenges

- New TTS evaluation metrics beyond MOS
- Multilingual TTS — one model, many Indic languages
- Indic prosody — schwa deletion, gemination, intonation
- Long-form TTS (audiobooks, news, education)

S2S

3 challenges

- Techniques to construct S2S datasets for Indic pairs
- Cascaded vs end-to-end S2S models
- Holistic S2S evaluation metrics

SQA

4 challenges

- Evaluation metrics for Spoken QA
- Cascaded vs end-to-end Spoken QA models
- Dataset creation techniques for Spoken QA
- MCQ vs descriptive Spoken QA

Speech LLMs

2 challenges

- Building Speech LLMs for Indic languages
- Speaker characteristics from Speech LLMs

Rich Data

2 challenges

- Rich transcription beyond plain text (prosody, disfluency, emotion)
- Data creation with minimal human effort

Shared Tasks: Language

Robust MT for Indian Languages

Move beyond aggregate BLEU. Diagnose where Indic MT actually breaks — by linguistic phenomenon, by dialect, by error type.

1

Challenge Set Evaluation

Linguistically grounded diagnostic test sets — minimal pairs targeting morphology, syntax, code-switching, idioms and domain usage — to isolate specific MT capabilities rather than aggregate-scoring everything.

2

Dialect Robustness

Evaluate MT on dialectally diverse inputs — Telangana Telugu, Kongu Tamil, informal/spoken variants and non-standard grammar. Tests whether systems generalise or silently normalise away dialect.

3

Error Analysis Track

Structured error taxonomy — morphological, syntactic, lexical, named-entity/transliteration, dialect misinterpretation. Submissions provide fine-grained analysis, not just outputs.

Strategic role: gives SLT-Eval a flagship text-side track and brings PLURAL Lab squarely into the workshop.

Workshop Host Trials

Two paths under active evaluation

Path A

IEEE Signal Processing Society

Mechanism

Technical Co-Sponsorship via SPS

Community

Strong signal-processing community fit; alignment with ICASSP/SLT cadence

Pros

- Prof. Vuppala chairs IEEE SPS Hyderabad Chapter (direct procedural access)
- Global Standard for Indic AI framing is directly inline with IEEE Standards
- IEEE SPS comfortably covers MT, NLG, LLM and multimodal tracks alongside speech

Cons

Heavier paperwork; Financial obligations and surplus-sharing rules are stricter

Status

Proposal to be submitted

Path B

ISCA

Mechanism

ISCA-supported Special Interest workshop

Community

Strong speech-research base; Interspeech adjacency

Pros

- Prof. Vuppala is an active member of ISCA
- ISCA-supported workshops have minimal procedural overhead
- Easier to position as a community-driven evaluation initiative rather than a sponsored conference

Cons

Less direct fit for MT, NLG and LLM tracks now planned in SLT-Eval

Status

Proposal to be submitted

Working preference: IEEE SPS Technical Co-Sponsorship, with ISCA retained as fallback.

Committee Footprint

International, multi-modality, multi-institution

General Chairs

- Anil Kumar Vuppala • IIIT Hyderabad (Lead)
- Radhika Mamidi • IIIT Hyderabad (Co-Lead)

Program Chairs

- Matt Post • Johns Hopkins HLTCOE
- Peter Bell • CSTR, Edinburgh
- Alan W Black • LTI, CMU
- Mathew Magimai-Doss • Idiap
- Dong Wang • CSLT, Tsinghua
- Lukáš Burget • BUT Speech@FIT

Steering Committee (advisory)

- Shrikanth Narayanan • USC SAIL
- Shinji Watanabe • CMU LTI; Chair, IEEE SPS SLP TC
- Simon King • Director, CSTR Edinburgh
- Honza Černocký • Head DCGM, BUT
- Thomas Fang Zheng • Director, CSLT Tsinghua
- Mikko Kurimo • Aalto
- Rajeev Sangal & Bayya Yegnanarayana • IIIT-H

Technical Committee & Local Organization

- Technical Committee: Khudanpur (JHU), Ganapathy (IISc), Bhiksha Raj (CMU), Hung-yi Lee (NTU), Yamagishi (NII), Bojar (Charles U.), Sitaram (MSR), Umesh & Mahadeva Prasanna, Sriram Ganapathy, Sakti, Kawahara, and 20+ more
- Local (IIIT-H LTRC): Dipti Misra Sharma, Manish Shrivastava, Parameswari Krishnamurthy, Chiranjeevi Yarra, Rajakrishnan Rajkumar, Vasudeva Varma, Arafat Ahsan, Anil Nelakanti, Prakash Yalla

Success Criteria — Status

Where each commitment stands today

CRITERION	STATUS	LABEL
Workshop branding and scope locked	Done	DONE
SLT-Eval 2027 website	Done	DONE
Call for Papers drafted across speech + NLP	Done	DONE
Shared tasks defined across speech and language tracks	In Progress	WIP
Committee shortlist drafted	In progress	WIP
IEEE SPS Technical Co-Sponsorship paperwork	In progress	WIP

Thank You

SLT-Eval 2027: Speech and Language Technology Evaluation Workshop

Any Questions?